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Abstract The report is about an exercise proposed by the edX translation UBCx Phot1xSilicon Photonics Design, Fabrication and Data Analysis.

1 Introduction

Silicon photonics enables the miniaturization of free-space optical systems by integrating optical components onto a silicon chip. This approach can be effectively applied to optical sensing technologies, which have become essential in modern engineering due to their ability to provide precise, distributed, and real-time measurements of physical parameters. Among these technologies, interferometric methods are particularly prominent. For instance, the Michelson interferometer operates by splitting a light beam into two separate paths, reflecting them back, and recombining them to generate interference patterns. These patterns contain detailed information about variations in optical path length caused by changes in distance, refractive index, or external perturbations such as strain and temperature. This same interferometric principle underpins optical frequency domain reflectometry, where the analysis of interference as a function of optical frequency enables spatially resolved sensing along an optical fiber.

2 Theory

2.1 Modeling

Beginning with the physics behind a y-branch splitter, let's consider an input intensity I_i and input electric field E_i. At the light is splitted at the output of this building-block, thus the intensity became $I_1 = I_2 = \frac{I_i}{2}$ and the electric field$I_1 = I_2 = \frac{I_i}{\sqrt{2}}$. For the combiner, we have $E_1 = \frac{E_{i_1}}{\sqrt{2}}$ and $E_{i_2} = \frac{E_i}{\sqrt{2}}$ which results $E_o = \frac{E_{i_1}+E_{i_2}}{\sqrt{2}}$

Considering a plane wave $E = E_0 e^{i(\omega t - \beta z)}$ with the propagation constant $\beta = \frac{2\pi n}{\lambda}$

$$E_{i_1} = E_1 e^{-i\beta_1 L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta_1 L_1} \tag{1}$$

$$E_{i_2} = E_2 e^{-i\beta_2 L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta_2 L_2} \tag{2}$$

Hence the output of the combiner:

$$E_o = \frac{E_{i_1} + E_{i_2}}{\sqrt{2}} = \frac{E_i}{2}(e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}) \tag{3}$$

And the intensity is:

$$I_o = \frac{I_i}{4}|e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}| \tag{4}$$

$$I_o = \frac{I_i}{4}(1 + cos(\beta \Delta L) \tag{5}$$

Using the siepic pdk from Luceda ipkiss, we ensure that the strip waveguide used has the following dimensions of width 500nm and heigh 220nm. Then we simulate the fundamental modes allowed to propagate in this waveguide, presented in the following figure 1 and 3 respectively the quasi-TE0 and quasi-TM0 fundamental modes.

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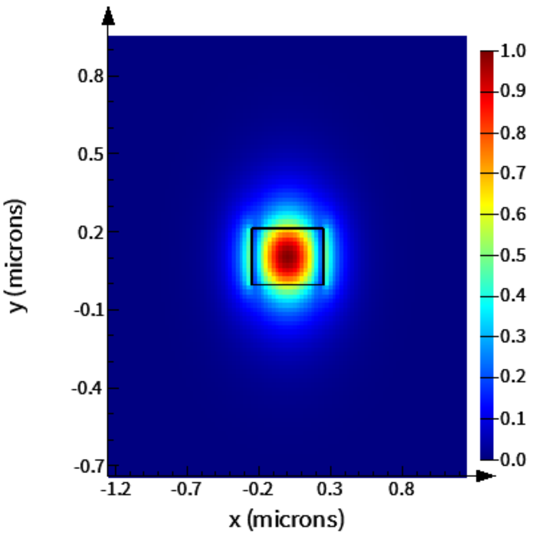

Fig. 1

TE mode in Si waveguide

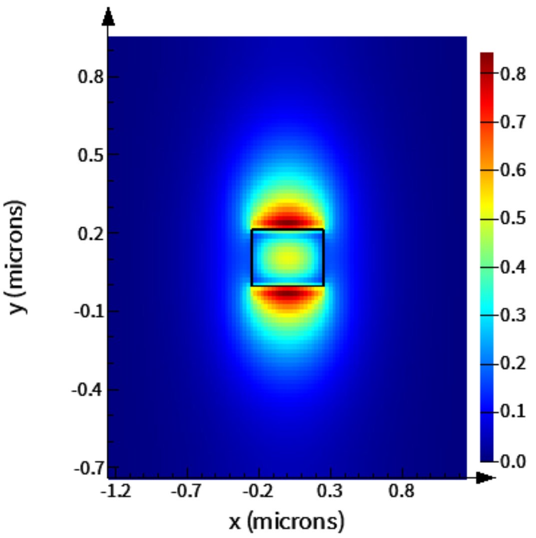

Fig. 2

TM mode in Si waveguide

2.1.1 effective index and group index versus wavelength

$$n_{eff}(\lambda_0) = 2.44 - 1.13 \cdot (\lambda - \lambda_0) - 0.06 \cdot (\lambda - \lambda_0)^2 \quad (6)$$

with $\lambda_0 = 1.55\ \mu m$

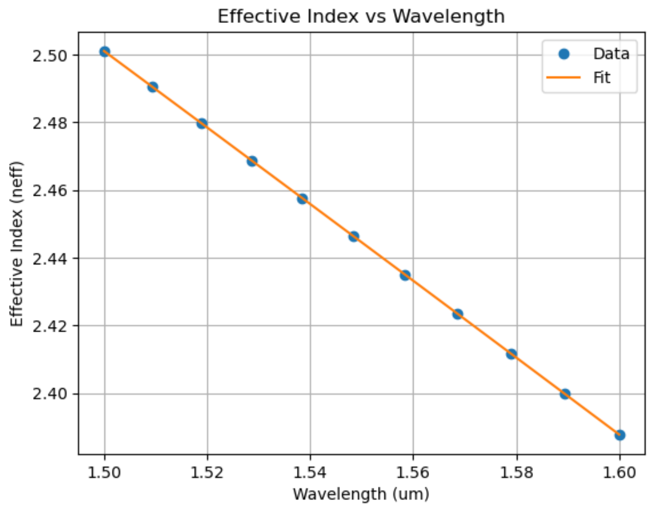

Fig. 3

linear regression of the effective index vs wavelength

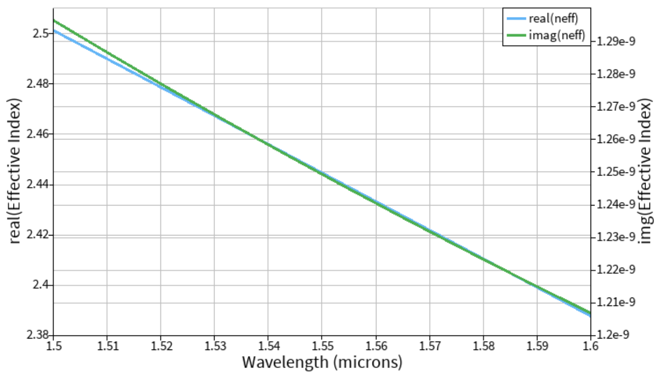

Fig. 4

effective index of quasi-TE mode function of wavelength

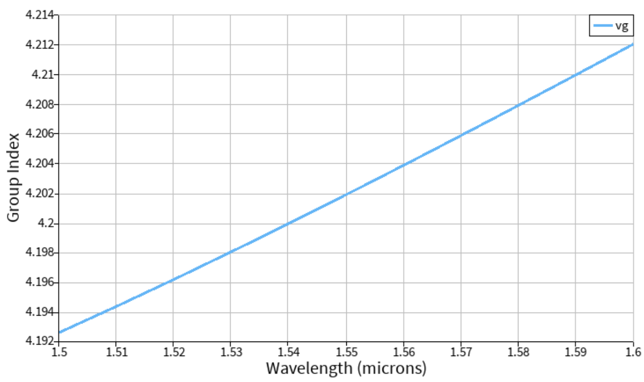

Fig. 5

group index of quasi-TE mode function of wavelength, $n_g = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda}$

2.1.2 Grating insertion loss

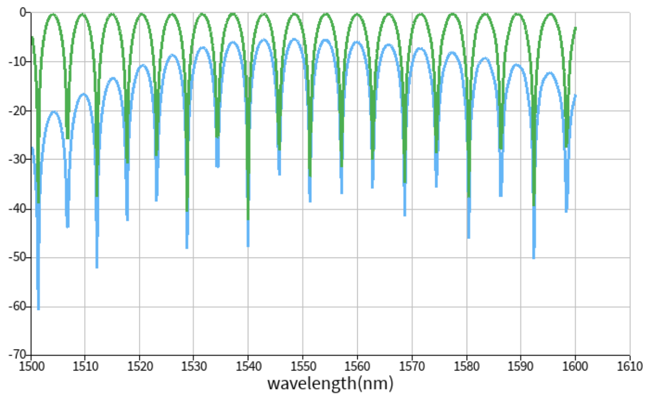

Fig. 6

MZI spectrum with (in blue) or without grating couplers (in green)

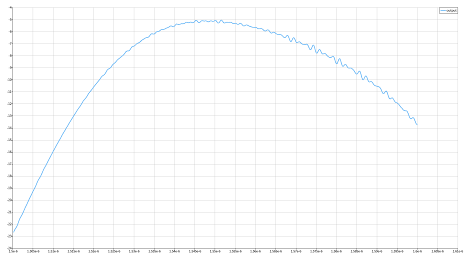

Fig. 7

Insertion loss of the grating coupler

2.1.3 Mach-Zehnder Interferometer (MZI)

unbalanced interferometer with identical wg

$$I_0 = \frac{I_i}{2}[1 + \cos(\beta \Delta L)] \tag{7}$$

In an MZI, light enters one port, splits, travels through the two arms once, and recombines at the output.
Physical length difference: $\Delta L = L_1 - L_2$

Total path difference: Since light only goes through the arms one time,$\Delta_{total} = \Delta L = L_1 - L_2$.

FSR Formula:

$$FSR_{MZI}(\lambda) = \frac{\lambda^2}{n_g \Delta L} \tag{8}$$

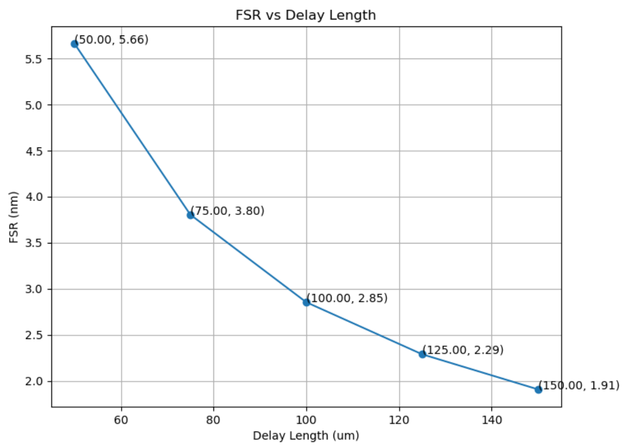

Fig. 8

FSR vs delay length for bend radius of 5um for MZI.

2.1.4 Michelson Interferometer

In a Michelson interferometer, light splits, travels down the arms, hits mirrors, and returns back through the same arms to recombine.
Physical length difference: $\Delta L = L_1 - L_2$.

Total path difference: Because light travels the arms twice (there and back), the effective difference is doubled: $\Delta_{total} = \Delta L = L_1 - L_2$.

FSR Formula:

$$FSR_{Michelson}(\lambda) = \frac{\lambda^2}{n_g 2\Delta L} \tag{9}$$

2.2 Simulations

2.2.1 Michelson Interferometer

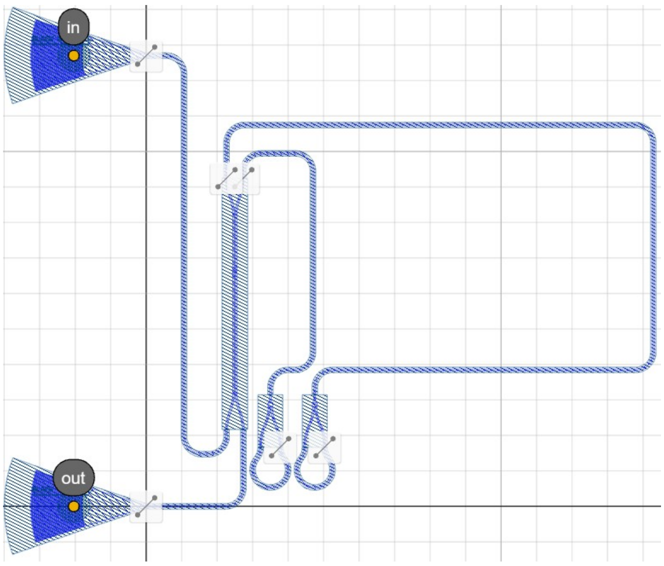

Fig. 9

Michelson Interferometer

2.2.2 Notes about the different layouts

layout **A**: Michelson interferometer with different length delay and a compact directional coupler. All component working at TE polarization for centered wavelength 1550 nm.
A WBG temperature sensor has been added to the layout for fun.

layout **B**: Michelson interferometer with different length delay and a adiabatic directional coupler. All component working at TE polarization for centered wavelength 1550 nm.

layout **C**: Attemp to make a Polarization-Splitter Rotator, inspired by the layout of this Polarization-SplitterRotator. The issues are that siepic pdk doesn't propose SiO2 cladding, necessary for this component to work and the other issue are pure layout missmatch between the width of the PSR's waveguides and the siepic waveguide section.
Tests structures of other devices has been added.

3 Fabrication

4 Experimental Data

5 Analysis

6 Conclusion

7 Acknowledgements

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